NIKHIL KHATRI

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in Python-based machine learning, deep learning, data preparation, model comparison and AI-enabled application development. Benchmarked 10+ ML algorithms for a classification problem, evaluated results using F1-score and accuracy, and built a CNN-based image-recognition system. Also experienced in Python, SQL, Pandas, NumPy, scikit-learn, TensorFlow, Keras and reproducible software development practices.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Java, Pandas, NumPy, Matplotlib
Machine Learning: Classification, regression, clustering, feature selection, model comparison, model evaluation, F1-score, accuracy
Deep Learning & Computer Vision: TensorFlow, Keras, CNNs, OpenCV, image preprocessing, model ensembling
GenAI & NLP: LangGraph, LangChain, Groq LLMs, Google Gemini API, intent classification, structured information extraction, prompt/output validation
Data & Backend: PostgreSQL, MongoDB, MySQL, FastAPI, SQLAlchemy, REST APIs
Engineering & Tools: Git, Maven, JUnit 5, Mockito, Docker, GitHub Actions, CI/CD, TDD

DATA SCIENCE & AI PROJECTS

THYROID DISEASE DETECTION — MACHINE LEARNING

Python | scikit-learn | Data Preparation | Classification

- Cleaned and prepared a large medical dataset and applied feature selection to build a classification workflow for thyroid disease detection.
- Benchmarked 10+ algorithms, including Logistic Regression, KNN, SVM, Decision Tree, Random Forest, Naive Bayes, DBSCAN, XGBoost and an Artificial Neural Network.
- Compared model performance using F1-score alongside accuracy, recognising the limitations of accuracy alone on imbalanced medical data.

REAL-TIME HAND SIGN RECOGNITION

Python | TensorFlow | Keras | CNN | OpenCV

- Built a real-time hand sign recognition system using convolutional neural networks (CNNs).
- Implemented image preprocessing and model ensembling to improve recognition results.

AI-FIRST CRM BACKEND — AGENTIC AI DATA EXTRACTION

Python | FastAPI | LangGraph | LangChain | PostgreSQL | SQLAlchemy

- Developed a six-tool LLM workflow for intent classification, structured data extraction, form editing, compliance validation, follow-up generation and summarization.
- Integrated validation before persistence and implemented REST endpoints with SQLAlchemy and PostgreSQL.

FITNESS MICROSERVICES PLATFORM

Java | Spring Boot | RabbitMQ | PostgreSQL | MongoDB | Docker

- Built a distributed application with six independently deployable Spring Boot services; integrated relational and NoSQL databases and asynchronous messaging.
- Applied automated build and deployment workflows using Docker and GitHub Actions.

EXPERIENCE

BUSINESS OPERATIONS | FAMILY RETAIL BUSINESS | Jhalawar, Rajasthan | Jul 2025 – Aug 2026

- Coordinated day-to-day retail operations, including inventory tracking, procurement, supplier coordination, sales reconciliation and customer interactions.
- Maintained financial records, tracked expenses and vendor payments, and supported customer relationship management and issue resolution.

EDUCATION & LEADERSHIP

B.Tech, Computer Science and Engineering | Silicon Institute of Technology, Silicon University | Dec 2021 – Jul 2025 | CGPA: 7.97/10

Joint Secretary, Silicon University Student Council — coordinated inter-departmental activities and student affairs; awarded Best Council Member (FY 2024).

CERTIFICATIONS

Deep Learning — IIT Ropar NPTEL (2025)
Generative AI With Keras, Pytorch And ChatGPT — Syllogistek Systems (2024)
Data Science, ML & Deep Learning Using Python — Syllogistek Systems (2023)
Full Stack PHP — Syllogistek Systems (2022)